# Chapter 1 — Java and the JVM

This chapter explains what Java is and what actually happens between typing `javac` (the Java compiler) and seeing output. You will set up a JDK (Java Development Kit — the compiler plus the other developer tools), write, **compile** (translate human-written source code into a form a machine can execute) and run programs three different ways, look inside a `.class` file, watch the Just-In-Time compiler (the part of the JVM that turns frequently run code into fast machine code while the program is running) and the garbage collector (the automatic cleaner that frees memory your program no longer uses) at work, experiment in `jshell` (an interactive prompt for typing Java line by line), and learn how Java versions and long-term-support releases (versions that vendors keep patching for many years) work, so you can choose the right JDK for a project.

**What you'll learn**

| # | Topic | You will be able to… |
|---|---|---|
| 1.1 | What Java is | explain Java, its history and where it is used in one minute |
| 1.2 | The Java ecosystem | tell Java SE, Jakarta EE, OpenJDK and vendor builds apart |
| 1.3 | JDK, JRE and JVM | say what each one contains and which one you install |
| 1.4 | Installing a JDK | install Java 25, set `JAVA_HOME`, verify the tools |
| 1.5 | Write → compile → run | build `HelloWorld`, read compiler errors, package a JAR |
| 1.6 | Bytecode and class files | read `javap -c` output and class-file versions |
| 1.7 | How the JVM runs code | describe class loading, the interpreter, JIT tiers and AOT caching |
| 1.8 | Memory and garbage collection | explain stack vs heap, reachability, generations and the collectors |
| 1.9 | jshell | try code instantly in the REPL |
| 1.10 | Launching source files | run `.java` files directly, including compact source files (Java 25) |
| 1.11 | IDEs and build tools | pick an editor and know where Maven/Gradle fit |
| 1.12 | Releases, LTS and JEPs | choose a Java version and read a feature's status |

All code lives in `code/`. It was compiled with `javac --release 25` (the book targets **Java 25 LTS**) and run on the JDK 26 installed on the author's machine; where output depends on the machine (timings, memory sizes) that is stated. For exam-level detail and traps on this material (class structure, `main` signatures, packages, garbage-collection eligibility), see the OCP repo's Chapter 1 — Building Blocks.

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## 1.1 What Java is

### What it is

Java is a **general-purpose, statically typed, object-oriented programming language** together with a **runtime platform** (the Java Virtual Machine) that runs compiled Java programs. A **runtime** is the software that must be present while a program is running to execute it and provide services such as memory management.

- *General-purpose*: not tied to one domain — the same language builds banking back-ends, Android apps (historically), big-data engines and command-line tools.
- *Statically typed*: every variable has a type known at compile time, so the compiler rejects `int n = "five";` before the program ever runs.
- *Object-oriented*: code is organised into **classes** (blueprints) and **objects** (instances built from them). Modern Java also supports functional style (lambdas — short, unnamed functions you can pass around as values; streams — pipelines that filter and transform collections of data — Chapters 11–12).

Analogy: Java source is like a recipe written in one standard language; the JVM is a professional kitchen installed in every restaurant (Windows, macOS, Linux, containers — lightweight, isolated packages such as Docker images that bundle an app with everything it needs to run). Write the recipe once and any kitchen can cook it. That is the origin of the slogan *"write once, run anywhere"*.

### A short history

| Year | Event |
|---|---|
| 1991 | James Gosling and a small team at Sun Microsystems start the "Green Project" for consumer devices. The language is first called **Oak**. |
| 1995 | Renamed **Java** and announced publicly (May 1995); browser "applets" (small Java programs embedded in web pages) make it famous. |
| 1996 | JDK 1.0 released (January 1996). |
| 2004 | J2SE 5.0: generics, enhanced `for`, autoboxing, enums, annotations, varargs — the biggest language update until Java 8. |
| 2006–07 | Sun open-sources the JDK as **OpenJDK** (GPL v2 — a free, open-source licence — with the Classpath Exception). |
| 2010 | Oracle acquires Sun and becomes Java's steward. |
| 2014 | Java 8: lambdas, streams, `java.time`. Still widely deployed. |
| 2017 | Java 9: the module system (JPMS — groups packages into named, sealed-off modules) and `jshell`. |
| 2018 | Java 10 starts the **six-month release cadence** (a new feature release every March and September). |
| 2021 | Java 17 LTS; long-term-support releases now every **two years**. |
| 2023 | Java 21 LTS: virtual threads, pattern matching for `switch`, record patterns. |
| 2025 | **Java 25 LTS** (16 Sept 2025): compact source files and instance `main`, flexible constructor bodies, module imports, scoped values. |
| 2026 | Java 26 (March) and Java 27 (15 Sept 2026) — non-LTS feature releases. |

Don't worry if feature names in this table (generics, enums, lambdas, virtual threads, records, pattern matching …) are new to you — each one is explained in its own later chapter.

### Design traits

| Trait | What it means in practice | Honest caveat |
|---|---|---|
| Platform independence | Compiled bytecode (the JVM's portable instruction format) runs on any OS/CPU that has a JVM. | You still need a JVM (or a native-image build — a program pre-compiled into a stand-alone executable) on the target. |
| Static typing | Type errors are caught by the compiler; IDEs (code editors with built-in tools) can refactor (restructure code without changing behaviour) safely. | More declarations than Python or JavaScript (softened by `var`, records). |
| Automatic memory management | A garbage collector frees unused objects; no `free()` / `delete`. | GC uses CPU and memory; you can still leak by keeping references (handles that point to objects). |
| Backward compatibility | Code compiled for Java 8 usually runs unchanged on Java 25. | Old APIs (the classes and methods a library exposes for you to call) linger (e.g. `Date`, `Vector`) and old habits persist. |
| Rich standard library (ready-made code shipped with Java) | Collections, concurrency (running several tasks at once), networking, HTTP client, `java.time`, crypto — all built in. | Large; learning which API is the modern one takes time. |
| Performance | The JIT compiler turns hot code (code that runs very often) into optimised machine code (instructions the CPU executes directly); long-running servers are fast. | Start-up and warm-up are slower than native languages (mitigated by AOT — ahead-of-time — caches, native images). |
| Safety | No pointer arithmetic (computing raw memory addresses by hand), array bounds are checked, bytecode is verified. | Checks have a (small) cost. |

### Where Java is used (real-world examples)

- **Enterprise back-ends** (the server-side part of an application that holds the business logic and data): banks, insurers, airlines and retailers run large Spring Boot / Jakarta EE services (popular **frameworks** — big reusable libraries that give an application its structure).
- **Big data and streaming**: Apache Kafka (moves streams of events between systems), Apache Cassandra (a distributed database), Elasticsearch (a search engine), Apache Hadoop and much of Apache Flink (large-scale data processing) run on the JVM (written in Java and Scala).
- **Developer tools**: IntelliJ IDEA, Eclipse, Maven, Gradle, Jenkins.
- **Android**: apps were historically written in Java (Android uses the Java *language* and much of its library, but its own runtime, ART, not the JVM); Google has recommended Kotlin first since 2019.
- **Games**: Minecraft: Java Edition.
- **Embedded and smart cards**: Java Card runs on SIM and bank cards.

Other languages also compile to JVM bytecode — **Kotlin, Scala, Groovy, Clojure** — and interoperate with Java libraries. Learning the JVM helps with all of them.

> **TL;DR**

Java = a statically typed, object-oriented language + a portable runtime (the JVM). It is mature (1996), backward compatible, fast after warm-up, and dominant in server-side and data infrastructure.

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## 1.2 The Java ecosystem: SE, EE, OpenJDK and vendors

### What it is

"Java" can mean several things. This table untangles them.

| Name | What it is |
|---|---|
| **Java SE** (Standard Edition) | The *specification* of the language, JVM and core library (`java.lang`, `java.util`, …). Each version (SE 25) is defined through the Java Community Process (the formal process through which companies and individuals agree on changes to Java). |
| **JDK** (Java Development Kit) | A concrete *implementation* of Java SE plus tools (`javac`, `jshell`, `jar`, `javadoc`, `jlink`…). |
| **OpenJDK** | The open-source (source code public and free to use and modify) project where the reference JDK is developed (openjdk.org). Almost every JDK you can download is built from it. |
| **Vendor builds** | Oracle JDK, Eclipse Temurin (Adoptium), Amazon Corretto, Azul Zulu, Microsoft Build of OpenJDK, Red Hat build of OpenJDK, BellSoft Liberica, SAP Machine. Same code base, different support/licensing and extras. |
| **Jakarta EE** (formerly Java EE) | Specifications for enterprise APIs (servlets — handle web requests; JPA persistence — save objects to databases; CDI — wires objects together…) run on top of Java SE. Moved from Oracle to the Eclipse Foundation (2017) and renamed. |
| **Java ME / Java Card** | Small-device and smart-card editions. |
| **GraalVM** | A JDK distribution from Oracle with the Graal compiler and **Native Image**, which compiles Java ahead of time into a standalone native executable (a program file the OS runs directly, with no JVM needed). |

#### Which vendor should I install?

For learning, any OpenJDK build of version 25 is fine; **Eclipse Temurin** is a common neutral choice. Companies pick a vendor by support contract, cloud (Corretto on AWS, Microsoft's build on Azure) or licensing. Oracle JDK 17 and later are released under the *Oracle No-Fee Terms and Conditions*, which allow free **production** use (running the real, live system that users depend on) for a limited period (for JDK 25, updates are free until September 2028), after which later updates need a paid licence; OpenJDK builds from other vendors are GPL-licensed.

### Common mistakes

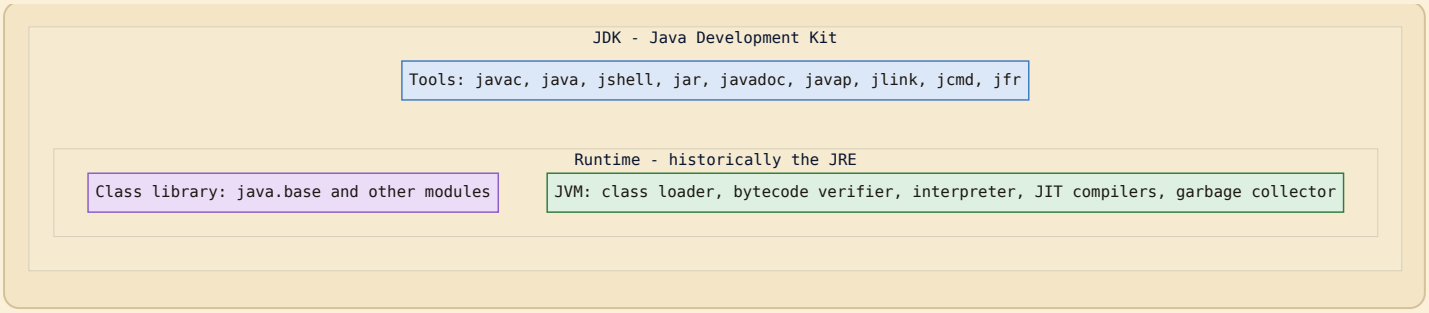
- Thinking "OpenJDK" is lower quality than "Oracle JDK" — they are built from the same source.
- Confusing Java with **JavaScript**; they share only a name and C-like syntax. (See the JavaScript Handbook.)

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## 1.3 JDK, JRE and JVM

### What it is

Three nested layers:




| Layer | Contains | Who needs it |
|---|---|---|
| **JVM** (Java Virtual Machine) | An abstract machine that loads bytecode, verifies it and executes it (interpreting and JIT-compiling), and manages memory. **HotSpot** is the JVM inside OpenJDK builds. | Every Java program at run time |
| **JRE** (Java Runtime Environment) | JVM + standard class library — enough to *run* programs, not compile them. | End users (historically) |
| **JDK** | JRE + development tools (`javac` compiler, `jshell`, `jar`, `javadoc`, debuggers, profilers). | Developers — **install this** |

*Virtual machine* here means a program that pretends to be a CPU with its own instruction set (bytecode). It is not a VM like VirtualBox that emulates a whole computer.

### Why the JRE "disappeared"

Since JDK 11, Oracle no longer ships a separate JRE download. Java 9's module system introduced `jlink`, which builds a **custom runtime** containing only the modules (named groups of packages, such as `java.base`) your application needs. Some vendors (for example Temurin) still publish JRE-only builds for convenience, mainly for containers.

A real example from this machine — a runtime containing only `java.base` is enough to run `HelloWorld`:

```
jlink --add-modules java.base --strip-debug --no-header-files --no-man-pages --output rt
du -sh rt                       # 44M   (the full JDK folder here is 377M)
rt/bin/java -cp out ch01.hello.HelloWorld
```

Output:

```
Hello, World!
Arguments received: 0
```

## In the interview

**Q: Difference between JDK, JRE and JVM?** The JVM executes bytecode and manages memory; the JRE is the JVM plus the standard library (enough to run programs); the JDK is the JRE plus development tools like `javac`. Developers install a JDK; production images often use a `jlink`-trimmed runtime.

- *Is the JVM platform-independent?* No — the **bytecode** is portable; each OS/CPU needs its own JVM build.
- *What is HotSpot?* The JVM implementation in OpenJDK, named for its strategy of finding "hot" code and compiling it.

## Self-check

1. You want to compile code. Is a JRE enough? → No, you need the JDK (`javac`).
2. What makes a Java program run on both Linux and Windows? → The same bytecode runs on each platform's own JVM.
3. What does `jlink` produce? → A trimmed runtime image containing only the modules you list.

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## 1.4 Installing a JDK and checking it works

### How it works

1. **Download** an OpenJDK 25 build (e.g. Temurin from adoptium.net) or use a version manager such as **SDKMAN!** (`sdk list java`, then `sdk install java <identifier>`) on macOS/Linux, or your OS package manager (a tool such as `apt` or Homebrew that installs software for you).
2. **Set** `JAVA_HOME` to the JDK folder and put `$JAVA_HOME/bin` on your `PATH`. Build tools (Maven, Gradle — programs that automate compiling, downloading libraries, testing and packaging) and IDEs read `JAVA_HOME`.
   - An **environment variable** is a named setting (like `JAVA_HOME=/path/to/jdk`) that the operating system passes to every program it starts. `PATH` is the environment variable listing the folders the shell searches when you type a command such as `java`.
   - macOS: `export JAVA_HOME=$(/usr/libexec/java_home -v 25)`
   - Linux: `export JAVA_HOME=/usr/lib/jvm/<jdk-folder>`
   - Windows: *System Properties → Environment Variables*, then add `%JAVA_HOME%\bin` to `Path`.
3. **Verify** in a new terminal (the command-line window where you type commands):

```
java -version
javac -version
```

Output (on the machine used for this book):

```
openjdk version "26" 2026-03-17
OpenJDK Runtime Environment (build 26+35-2893)
OpenJDK 64-Bit Server VM (build 26+35-2893, mixed mode, sharing)
javac 26
```

*"mixed mode"* means the JVM both interprets and JIT-compiles; *"sharing"* means it uses Class Data Sharing (a pre-parsed archive of core classes that speeds start-up).

**Compiling for an older version: `--release`**

A newer JDK can produce class files for an older Java version: `javac --release 25 …` checks your code against the Java 25 language **and** the Java 25 API, so you cannot accidentally use a Java 26 method. That is how this book's code targets Java 25 while running on JDK 26. (The older `-source`/`-target` pair does not check the API — prefer `--release`.)

### Common mistakes

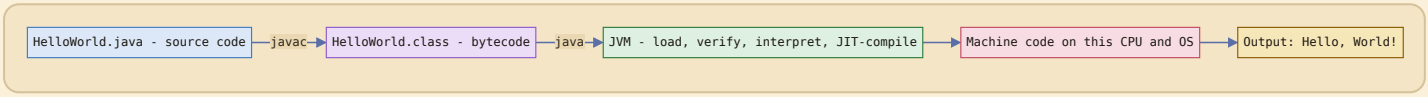
- Having several JDKs installed and `java` on `PATH` pointing at a different one than `javac` — check both versions.
- Setting `JAVA_HOME` to the `bin` folder instead of the JDK root.
- Installing only a JRE and then wondering why `javac` is "command not found".

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## 1.5 Write → compile → run

### What it is

Java is **compiled in two stages**: `javac` turns source code (`.java`) into platform-neutral **bytecode** (`.class` files); then the `java` launcher starts a JVM that loads and runs that bytecode.



### The program

`code/HelloWorld.java`:

```java
package ch01.hello;

/** The classic first program: one public class, one entry point. */
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
        System.out.println("Arguments received: " + args.length);
    }
}
```

| Part | Meaning |
|---|---|
| `package ch01.hello;` | The **package** (namespace) this class belongs to. It must be the first statement. The fully qualified class name becomes `ch01.hello.HelloWorld`. Packages are covered in Chapter 5. |
| `/** … */` | A **Javadoc comment**; the `javadoc` tool turns these into HTML API docs. `//` is a line comment, `/* … */` a block comment. |
| `public class HelloWorld` | Declares a class. A file may contain at most one **public** top-level class, and the file must be named after it (`HelloWorld.java`). |
| `public static void main(String[] args)` | The **entry point** the launcher calls. `public` = callable from anywhere; `static` = no object needed; `void` = returns nothing; `args` = command-line arguments (Chapter 4). |
| `System.out.println(...)` | Prints a line to **standard output** (the program's normal text output — the terminal, by default). `System` is a class in `java.lang` (imported automatically — an *import* lets you use a class by its short name), `out` is its static `PrintStream` field (a variable belonging to the class itself, holding an object that writes text). |
| `;` and `{ }` | Every statement ends with a semicolon; braces group statements into blocks. Java ignores indentation, but you should not. |

Since Java 25 a class may also declare a simpler `void main()` (no `static`, no parameters) — see 1.10. Traditional `public static void main(String[] args)` remains the standard for applications and works in every version.

### How to compile and run

```
javac -d out HelloWorld.java            # compile; -d puts classes in out/ch01/hello/
java -cp out ch01.hello.HelloWorld       # run: class path + fully qualified class name
java -cp out ch01.hello.HelloWorld a b c
```

Output:

```
Hello, World!
Arguments received: 0
Hello, World!
Arguments received: 3
```

1. `javac` parses the file (reads the text and works out its grammatical structure), checks types and writes `out/ch01/hello/HelloWorld.class` (the folder structure mirrors the package).
2. `java` starts a JVM; `-cp` (class path) tells it where to look for classes.
3. The JVM loads `ch01.hello.HelloWorld`, finds `main`, and calls it with the arguments.

### Reading errors: compile-time vs run-time

Compile-time errors are reported by `javac` with file, line and a caret. Real messages:

```
Hello.java:3: error: ';' expected
        System.out.println("hi")
                                ^
```

```
Hello.java:1: error: class Greeting is public, should be declared in a file named Greeting.java
```

```
Hello.java:3: error: incompatible types: String cannot be converted to int
        int n = "five";
                ^
```

Launcher errors happen when `java` cannot find the class:

| Command | Error | Fix |
|---|---|---|
| `java hello` | `NoClassDefFoundError: hello (wrong name: Hello)` on macOS/Windows (case-insensitive file systems); `ClassNotFoundException: hello` on Linux | Class names are case-sensitive. |
| `java Hello.class` | `ClassNotFoundException: Hello.class` | Pass the class **name**, not the file name. |
| `java HelloWorld` (class is in a package) | `Could not find or load main class HelloWorld` | Use the fully qualified name and the right `-cp`. |

### Packaging a JAR

A **JAR** (Java ARchive) is a ZIP file of classes plus a manifest (a small text file of metadata, i.e. data describing the archive). With a `Main-Class` entry it becomes runnable:

```
jar --create --file hello.jar --main-class ch01.hello.HelloWorld -C out .
java -jar hello.jar one two
```

Output:

```
Hello, World!
Arguments received: 2
```

Real projects let Maven or Gradle build JARs (Chapter 18).

## Common mistakes

- File name ≠ public class name (case matters: `helloworld.java` fails for `HelloWorld`).
- Forgetting `-cp`/wrong directory, or running from inside the package folder.
- Misspelling `main`'s signature (e.g. `String args` without `[]`) in a traditional class — it compiles, but the launcher reports no suitable main method.

## Self-check

1. What does `javac` produce? → `.class` files containing bytecode.
2. Why does `java Hello.class` fail? → The launcher wants a class name, not a file path.
3. Where does the compiled `ch01.hello.HelloWorld` go with `-d out`? → `out/ch01/hello/HelloWorld.class`.

## TL;DR

`javac` → bytecode; `java -cp <dir> <package.Class>` → JVM runs it. One public top-level class per file, named like the file.

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## 1.6 Bytecode and class files

### What it is

**Bytecode** is the instruction set of the JVM: compact, one-byte opcodes (operation codes — numbered instructions such as `iload`, `iadd`, `invokevirtual` …) for a **stack machine** — instructions push values onto an operand stack and pop them to compute. A `.class` file holds one class's bytecode plus a **constant pool** (names, strings, method references) and metadata.

### Why it matters

- It is the unit of portability: the same `.class` runs on every JVM.
- The JVM verifies it before running (no forged pointers, no operand-stack overflow or underflow, no jumps into the middle of an instruction from malformed code).
- Tools (profilers — measure where time and memory go; mocking libraries — create fake objects for tests; frameworks like Spring and Hibernate) read or generate bytecode at run time.

### Looking inside with `javap`

`code/Calculator.java`:

```
package ch01.bytecode;

public class Calculator {
    static int add(int a, int b) {
        return a + b;
    }

    public static void main(String[] args) {
        int total = add(2, 3);
        System.out.println("2 + 3 = " + total);
    }
}
```

`javap -c -cp out ch01.bytecode.Calculator` (a disassembler — turns bytecode back into readable instructions — shipped with the JDK):

```
public class ch01.bytecode.Calculator {
  public ch01.bytecode.Calculator();
    Code:
         0: aload_0
         1: invokespecial #1                  // Method java/lang/Object."<init>":()V
         4: return

  static int add(int, int);
    Code:
         0: iload_0
         1: iload_1
         2: iadd
         3: ireturn

  public static void main(java.lang.String[]);
    Code:
         0: iconst_2
         1: iconst_3
         2: invokestatic  #7                  // Method add:(II)I
         5: istore_1
         6: getstatic     #13                 // Field java/lang/System.out:Ljava/io/PrintStream;
         9: iload_1
        10: invokedynamic #19,  0             // InvokeDynamic #0:makeConcatWithConstants:(I)Ljava/lang/String;
        15: invokevirtual #23                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
        18: return
}
```

How to read it:

1. `javac` added a **default constructor** (a constructor is the special method that sets up a new object; this one is `Calculator()`) that calls `Object`'s constructor — you did not write it.
2. `add`: `iload_0`/`iload_1` push the two `int` parameters, `iadd` pops both and pushes the sum, `ireturn` returns it. (`i` = int.)
3. `main`: pushes constants 2 and 3, calls `add` (`(II)I` = takes two ints, returns int), stores the result in local slot 1.
4. String concatenation compiles to `invokedynamic … makeConcatWithConstants` (since Java 9), which lets the JVM choose the fastest concatenation strategy at run time.

### Class-file version

Every class file starts with the magic number `CAFEBABE` (fixed bytes that identify the file type), then a **major version** that says which Java produced it: major = 44 + Java version.

| Java | 8 | 11 | 17 | 21 | 25 | 26 |
|---|---|---|---|---|---|---|
| Major version | 52 | 55 | 61 | 65 | **69** | 70 |

```
javap -v -cp out ch01.bytecode.Calculator | grep major   # major version: 69
xxd out/ch01/bytecode/Calculator.class | head -1          # cafe babe 0000 0045 ...  (0x45 = 69)
```

An older JVM refuses newer class files with `UnsupportedClassVersionError` ("compiled by a more recent version of the Java Runtime (class file version 69.0), this version … only recognizes class file versions up to 65.0"). Fix: run on a newer JVM or compile with `--release <older>`.

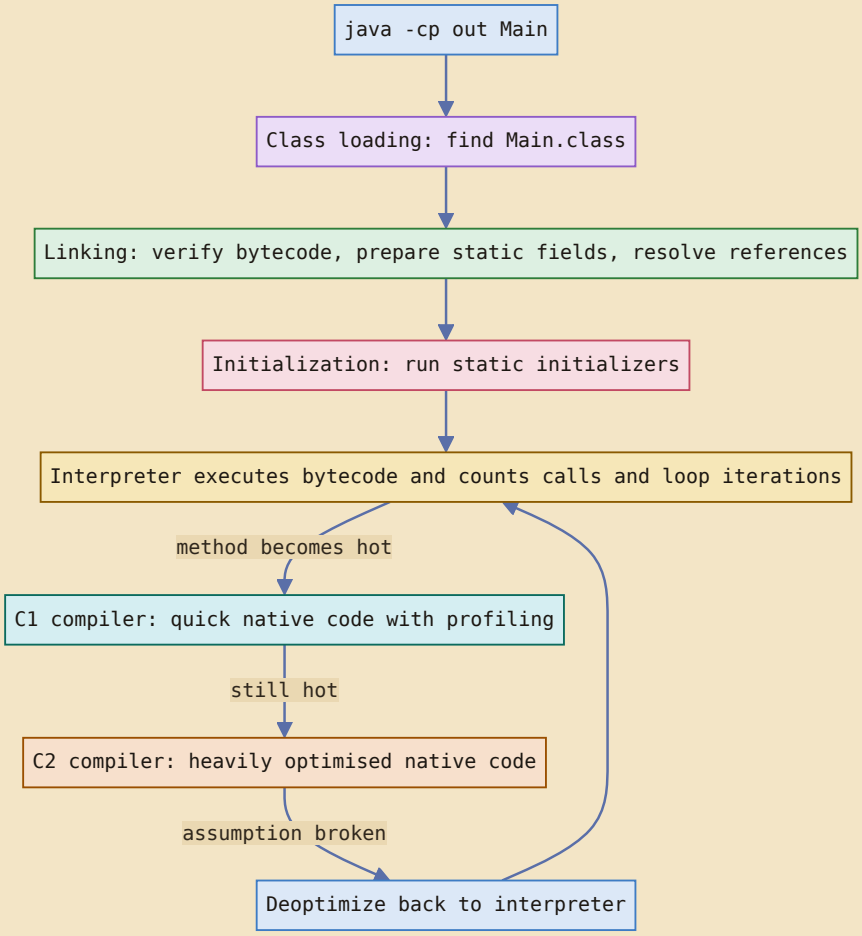
### Self-check

1. What kind of machine does bytecode target? → A stack-based virtual machine.
2. You get `UnsupportedClassVersionError … 69.0`. Which Java compiled it? → Java 25.
3. Who wrote the `Calculator()` constructor? → The compiler (default constructor).

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## 1.7 How the JVM runs your code: class loading, interpreter, JIT

### How it works




1. **Class loading.** *Class loaders* find `.class` bytes and turn them into classes, lazily (the first time a class is used). There are three built-in loaders: **bootstrap** (core `java.base` classes), **platform** (other JDK modules) and **application/system** (your class path). Each asks its parent first (*parent delegation*), so you cannot replace `java.lang.String` with your own.
2. **Linking.** *Verification* checks the bytecode is type-safe and well-formed; *preparation* allocates static fields with default values; *resolution* turns symbolic names (e.g. `java/io/PrintStream.println`) into direct references (done lazily).
3. **Initialization.** Static field initializers and `static { }` blocks run, once per class.
4. **Execution.** The **interpreter** runs bytecode instruction by instruction — starts instantly but is slow. It keeps counters of how often each method and loop runs.
5. **JIT compilation.** *Just-In-Time* compilation converts **hot** methods into native machine code while the program runs. HotSpot uses **tiered compilation** (default since Java 8):

| Tier | Who | What |
|---|---|---|
| 0 | Interpreter | Runs bytecode, gathers counts |
| 1–3 | **C1** ("client") compiler | Compiles fast with light optimisation; tier 3 adds profiling (which types and branches actually occur) |
| 4 | **C2** ("server") compiler | Uses the profile for aggressive optimisation: inlining, loop unrolling, escape analysis, dead-code removal |

Those optimisations in plain words: **inlining** copies a small method's body into its caller to skip the call; **loop unrolling** repeats a loop body several times per iteration to cut loop overhead; **escape**

**analysis** spots objects that never leave a method so they need not live on the heap; **dead-code removal** deletes work whose result is never used.

6. **Speculation and deoptimization.** C2 optimises for what it has *seen* (e.g. "this call always hits `ArrayList`"). If that assumption later breaks, the JVM throws away the compiled code ("uncommon trap") and falls back to the interpreter, then may recompile.

### Watching the JIT

`code/HotLoop.java` calls a tiny method five million times:

```
static long square(long x) { return x * x; }

public static void main(String[] args) {
    long sum = 0;
    for (long i = 0; i < 5_000_000; i++) {
        sum += square(i % 1000);
    }
    System.out.println("sum = " + sum);
}
```

`java -XX:+PrintCompilation -cp out ch01.jit.HotLoop | grep -E "HotLoop|sum ="` (real output; timestamps vary by machine):

```
15    9       3       ch01.jit.HotLoop::square (4 bytes)
15   10       4       ch01.jit.HotLoop::square (4 bytes)
15    9       3       ch01.jit.HotLoop::square (4 bytes)   made not entrant: not used
16   11 %     3       ch01.jit.HotLoop::main @ 4 (43 bytes)
16   12       3       ch01.jit.HotLoop::main (43 bytes)
16   13 %     4       ch01.jit.HotLoop::main @ 4 (43 bytes)
16   11 %     3       ch01.jit.HotLoop::main @ 4 (43 bytes)   made not entrant: OSR invalidation of lower level
19   13 %     4       ch01.jit.HotLoop::main @ 4 (43 bytes)   made not entrant: uncommon trap
sum = 1664167500000
```

Columns: milliseconds since start, compile ID, flags, **tier**, method. `square` was compiled by C1 (tier 3) and within the same millisecond by C2 (tier 4), and the C1 version was discarded ("made not entrant"). `%` marks **on-stack replacement (OSR)**: `main` runs only once, but its loop is hot, so the JVM compiles the loop and switches to it *while the method is still running*. The final line is a deoptimization.

### Start-up and warm-up: AOT caching

Because the JVM starts in the interpreter, a Java service is slower for its first seconds (**warm-up**). Recent JDKs attack this with an **ahead-of-time (AOT) cache** recorded from a training run:

- JDK 24 (JEP 483 — a *JEP* is a numbered JDK Enhancement Proposal, see 1.12): classes can be loaded and linked ahead of time into a cache.
- JDK 25 (JEP 514): one-step cache creation, `java -XX:AOTCacheOutput=app.aot …` on a training run, then `java -XX:AOTCache=app.aot …` in production.
- JDK 25 (JEP 515): the cache also stores method profiles, so the JIT can start optimising sooner.

Separately, **GraalVM Native Image** compiles the whole application to a native executable before it runs: very fast start-up and small memory, at the cost of long builds and restrictions on reflection (code inspecting classes and calling methods by name at run time) and dynamic class loading.

### Trade-offs: interpreted vs JIT vs AOT

| Approach | Start-up | Peak speed | Portability | Example |
|---|---|---|---|---|
| Pure interpretation | Instant | Slow | High | Early JVMs, CPython (the standard Python interpreter) |
| JIT (HotSpot) | Slower (warm-up) | Very high — uses live profiles | High (bytecode) | Normal `java` |
| AOT cache (JDK 24+) | Faster | Same as JIT | High | `-XX:AOTCache` |
| Native image | Very fast | Good, often a bit lower without profile data | One binary per OS/CPU | GraalVM `native-image` |

### In the interview

**Q: Is Java compiled or interpreted?** Both. `javac` compiles source to bytecode ahead of time; the JVM then interprets that bytecode and JIT-compiles hot methods to native code at run time, using profiling data. That is why long-running Java services get faster after warm-up.

- *What is tiered compilation?* Interpreter → C1 (fast compile, profiling) → C2 (optimising compile using the profile).
- *What is deoptimization?* Discarding compiled code whose speculative assumption became false and returning to the interpreter.
- *Why is a microbenchmark with a simple timing loop misleading?* It mixes interpreted, C1 and C2 phases and the JIT may remove "useless" work; use JMH (Java Microbenchmark Harness, a tool for accurate small-scale timing; Chapter 16).

### Self-check

1. When is a class loaded? → Lazily, the first time it is actively used.
2. What does `%` mean in `PrintCompilation` output? → On-stack replacement of a hot loop.
3. Why can a JIT beat a static compiler sometimes? → It optimises for the types and branches actually seen at run time.

---

## 1.8 Memory and garbage collection — an overview

This section gives the picture you need now; algorithms, tuning and profiling are in Chapter 16.

### What it is

The JVM divides memory into areas:

> **Key term — thread:** a thread is one independent sequence of instructions running inside a program. A program can have many threads running at the same time (for example, one per incoming web request), and each has its own call stack but shares the heap with the others. Threads are covered properly in the concurrency chapters.

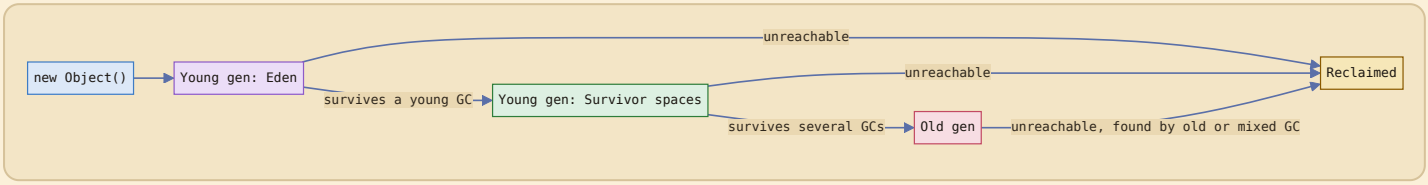
| Area | Holds | Lifetime |
|---|---|---|
| **Stack** (one per thread) | *Frames* for each method call: local variables, parameters, partial results. Primitive locals live here; for objects, only the **reference** (a pointer-like handle) lives here. | Frame disappears when the method returns |
| **Heap** (shared) | All objects and arrays (`new …`). | Until the garbage collector reclaims them |
| **Metaspace** | Class metadata (methods, field layouts), in native memory (memory the JVM takes from the OS outside the Java heap). | While the class loader is alive |
| **Code cache** | JIT-compiled machine code. | Until evicted/deoptimized |

**Garbage collection (GC)** is automatic reclamation of heap objects that are no longer **reachable**. An object is reachable if a chain of references leads to it from a **GC root** — a local variable in an active frame, a static field, an active thread and a few JVM-internal references. Anything not reachable is garbage, even if it still points to other objects (so reference cycles are collected too).

Analogy: a stadium cleaner removes any seat cushion no ticket-holder is still attached to — they do not ask the cushion whether it is needed.

## How it works (the common core)

1. **Mark**: starting from the roots, follow references and mark every reachable object.
2. **Sweep / compact / copy**: free the unmarked memory; many collectors also move live objects together to avoid fragmentation (free memory scattered in many small gaps that are too small to use).
3. **Generations**: most objects die young (the *weak generational hypothesis* — think temporary strings in a request). Generational collectors therefore split the heap into a **young generation**, collected often and cheaply, and an **old generation** for survivors, collected rarely.
4. **Pauses**: some work needs application threads stopped (**stop-the-world** pause); modern collectors do most work **concurrently** (at the same time as) with your threads.




### The collectors in JDK 25

| Collector | Flag | Design | Good for |
|---|---|---|---|
| **G1** (Garbage-First) | `-XX:+UseG1GC` | Generational, region-based (heap split into many equal-sized chunks), mostly concurrent marking; aims for a pause-time goal (default 200 ms) | **The default** — general-purpose servers and desktops |
| **ZGC** | `-XX:+UseZGC` | Concurrent, generational (the only mode since JDK 24); pauses typically under a millisecond regardless of heap size | Large heaps, latency-sensitive services (where the time each request takes matters) |
| **Shenandoah** | `-XX:+UseShenandoahGC` | Concurrent compacting; generational mode became a product feature in JDK 25 | Low latency; included in many OpenJDK builds (not Oracle JDK) |
| **Parallel** | `-XX:+UseParallelGC` | Stop-the-world, many threads; maximises throughput (total work done per second) | Batch jobs where pauses do not matter |
| **Serial** | `-XX:+UseSerialGC` | Single-threaded, stop-the-world | Tiny heaps, small containers |
| **Epsilon** | experimental | Allocates, never collects | Testing and very short-lived jobs |

The JVM picks G1 unless the machine looks "small" (fewer than 2 CPUs or under 1792 MB of memory), where it chooses Serial; JDK 27 (JEP 523) makes G1 the default everywhere. By default the maximum heap is ¼ of physical memory — `WhereAmI` below reports 12 288 MiB on a 48 GiB machine (MiB/GiB are binary units: 1 MiB = 1024 × 1024 bytes, 1 GiB = 1024 MiB). **Latency** and **throughput** are system-design terms you will meet again; see System Design Fundamentals.

### Watching the GC

`code/GarbageMaker.java` creates 200 000 short-lived 1 KiB arrays and keeps ten 1 MiB arrays:

```
List<byte[]> keep = new ArrayList<>();          // long-lived: stays reachable
for (int i = 0; i < 200_000; i++) {
    byte[] temp = new byte[1024];                // short-lived: garbage after this iteration
    temp[0] = (byte) i;
    created++;
    if (i % 20_000 == 0) keep.add(new byte[1024 * 1024]);
}
```

`java -Xlog:gc -Xmx256m -cp out ch01.gc.GarbageMaker` (selected lines from one real run; timings and the exact sequence of collections vary from run to run):

```
[0.004s][info][gc] Using G1
[0.017s][info][gc] GC(0) Pause Young (Concurrent Start) (G1 Humongous Allocation) 1M->1M(10M) 0.407ms
[0.017s][info][gc] GC(1) Concurrent Mark Cycle
[0.018s][info][gc] GC(2) Pause Young (Normal) (G1 Evacuation Pause) 6M->3M(10M) 0.195ms
[0.022s][info][gc] GC(4) Pause Young (Prepare Mixed) (G1 Evacuation Pause) 22M->5M(30M) 0.580ms
[0.024s][info][gc] GC(5) Pause Young (Mixed) (G1 Evacuation Pause) 22M->5M(30M) 0.305ms
...
temporary arrays created: 200000
arrays still reachable  : 10
```

Reading a line: `22M->5M(30M) 0.580ms` = heap in use before → after the collection (current heap size), and the pause length. Young collections free most memory because the temporary arrays are already dead. "Humongous" means an object at least half a G1 region in size (the 1 MiB arrays) that G1 allocates specially. With `-XX:+UseZGC` the first line reads `Using The Z Garbage Collector`.

Ask the JVM about itself with `code/WhereAmI.java`:

```
Runtime.Version v = Runtime.version();
System.out.println("Feature release : " + v.feature());
System.out.println("class file ver. : " + System.getProperty("java.class.version"));
System.out.println("Max heap (MiB)  : " + Runtime.getRuntime().maxMemory() / (1024 * 1024));
```

Output (this machine; yours will differ):

```
Feature release : 26
java.vendor     : Oracle Corporation
java.vm.name    : OpenJDK 64-Bit Server VM
class file ver. : 70.0
os.name         : Mac OS X
CPU cores seen  : 14
Max heap (MiB)  : 12288
```

Note the program was *compiled* for Java 25 but *runs* on JDK 26 — newer JVMs run older class files.

### Common mistakes

- Believing Java cannot leak memory. A static `Map` cache that only grows keeps every entry reachable forever — a **memory leak**.
- Calling `System.gc()` to "free memory": it is only a request, and usually hurts.
- Relying on `finalize()` for cleanup: deprecated for removal (JEP 421, Java 18). Use try-with-resources (Chapter 8).
- Setting `-Xmx` (the flag for the maximum heap size) larger than a container's memory limit (the JVM reads container limits automatically; let it, or size explicitly).

### In the interview

**Q: How does Java garbage collection decide what to free?** By reachability, not reference counting (keeping a count of how many references point at each object): anything not reachable from GC roots (thread stacks, static fields, JNI handles — references held by native C code via the Java Native Interface) is garbage, so cycles are collected. Most collectors are generational because most objects die young.

- *Default GC?* G1 since Java 9 on server-class machines.
- *When would you use ZGC?* Large heaps or strict latency targets, where sub-millisecond pauses matter more than a little throughput.
- *Stack vs heap?* Stack: per-thread frames with locals, freed on return. Heap: shared objects, freed by GC.

### Self-check

1. Two objects reference each other but nothing else references them. Are they collected? → Yes; they are unreachable.
2. Where does `int x = 5;` inside a method live? Where does `new int[5]` live? → `x` on the stack; the array on the heap (the reference to it on the stack).
3. Why are young collections cheap? → Most young objects are dead, and the collector only copies the few live ones.

### TL;DR

Stack = method frames; heap = objects. The GC frees unreachable objects; generational collectors exploit "most objects die young". Default is G1; ZGC/Shenandoah for low latency; Parallel for throughput.

---

## 1.9 jshell — the Java REPL

### What it is

`jshell` (Java 9, JEP 222) is a **REPL** — *Read-Evaluate-Print Loop*: type a snippet, see the result immediately, no class or `main` needed. Ideal for trying an API or checking an operator's behaviour.

#### A session

Real session (lines after `jshell>` are what was typed):

```
|  Welcome to JShell -- Version 26
|  For an introduction type: /help intro

jshell> int x = 21
x ==> 21

jshell> x * 2
$2 ==> 42

jshell> "Java".length()
$3 ==> 4

jshell> String shout(String s) { return s.toUpperCase() + "!"; }
|  created method shout(String)

jshell> shout("hi")
$5 ==> "HI!"

jshell> /vars
|    int x = 21
|    int $2 = 42
|    int $3 = 4
|    String $5 = "HI!"
```

- Expressions not assigned to a variable get scratch names (`$2`, `$3`…) you can reuse.
- Semicolons are optional for single statements.
- Methods can be declared at the top level and redefined.

| Command | Does |
|---|---|
| `/help`, `/help intro` | Help |
| `/vars`, `/methods`, `/types`, `/imports` | List what you have defined/imported |
| `/list` | Show your snippets |
| `/edit <name>` | Open a snippet in an editor |
| `/save file.jsh`, `/open file.jsh` | Save/load a session |
| `/reset`, `/exit` | Clear state, quit |
| Tab | Completion and method signatures |

On the JDK 26 used here, `/imports` shows `import java.base`, so every public class in the `java.base` module is usable without imports.

### When not to use it

For anything you want to keep, test or share, write a source file. jshell is a scratchpad.

---

## 1.10 Launching source files directly

### Single-file and multi-file source launch

Since **Java 11** (JEP 330) the launcher can compile a single `.java` file in memory and run it: `java Hello.java`. No `.class` files are written. Since **Java 22** (JEP 458) the launched program may use other source files in the same directory tree; they are compiled on demand.

`code/launch/Main.java` uses `code/launch/Greeter.java`:

```java
// Main.java
public class Main {
    public static void main(String[] args) {
        System.out.println(Greeter.greet("launcher"));
    }
}
// Greeter.java
public class Greeter {
    static String greet(String who) {
        return "Hello from " + who + ", compiled in memory";
    }
}
```

```
cd code/launch && java Main.java
```

Output:

```
Hello from launcher, compiled in memory
```

On Linux/macOS a single-file program can even start with a **shebang** line (a first line starting with `#!` that tells Unix which program should run the file) (`#!/usr/bin/env -S java --source 25`) in a file *without* the `.java` extension and be run as a script.

### Compact source files and instance `main` — Java 25

JEP 512 (final in **Java 25**, after previews — opt-in trial versions, see 1.12 — in 21–24) removes the ceremony for small programs:

- A **compact source file** may declare methods and fields with no enclosing class; the compiler wraps them in an implicit, unnamed-to-you `final` class (a class that cannot be extended).
- `main` may be an **instance method** (a method that runs on an object rather than on the class itself) with no parameters: `void main()`.
- A compact source file automatically imports every public class of the `java.base` module (as if it began with `import module java.base;`), so `List`, `Map`, `Path` … need no imports.
- The new class `java.lang.IO` offers `IO.println(…)`, `IO.print(…)`, `IO.println()` and `IO.readln(prompt)`/`IO.readln()` for simple console I/O (input/output — reading what the user types and printing text; built on `System.out` and `System.in`).

`code/compact/HelloCompact.java` — the whole file:

```
void main() {
    List<String> langs = List.of("Java", "Kotlin", "Scala");   // java.util.List without an import
    IO.println("Hello from a compact source file!");
    IO.println("JVM languages: " + langs);
}
```

```
cd code/compact && java HelloCompact.java
```

Output:

```
Hello from a compact source file!
JVM languages: [Java, Kotlin, Scala]
```

How the launcher chooses `main`: if the class has a `main(String[])`, that is used; otherwise a no-argument `main()`. If the chosen method is not `static`, the launcher first creates an instance with the no-argument constructor.

### Trade-offs

| Style | Best for | Limits |
|---|---|---|
| `javac` + `java` | Real applications, anything with packages or dependencies | Two steps |
| `java File.java` | Scripts, examples, teaching, small tools | Compiles every run; no build-tool dependency management |
| Compact source file | First programs, quick utilities | No `package`; cannot be referenced by name from other code; grows into a normal class as the program grows |

### Common mistakes

- Adding a `package` declaration to a compact source file (not allowed).
- Assuming instance `main` works on Java 21 without `--enable-preview` — it was preview until Java 25.

---

## 1.11 IDEs, editors and build tools

### What they are

An **IDE** (Integrated Development Environment) combines an editor, compiler integration, debugger (a tool that pauses a running program so you can step through it and inspect variables), refactoring and test runner.

| Tool | Notes |
|---|---|
| **IntelliJ IDEA** (JetBrains) | The most popular Java IDE; strong refactoring and inspections. Free and paid tiers. |
| **Eclipse IDE** | Free, open source, long-standing; its own incremental compiler. |
| **VS Code** + Microsoft's *Extension Pack for Java* | Lightweight; good for mixed-language work. |
| **Apache NetBeans** | Free, open source. |

**Build tools** automate compiling, dependency download, testing and packaging: **Maven** (XML `pom.xml`, fixed lifecycle) and **Gradle** (Groovy/Kotlin DSL — a small domain-specific language for writing build scripts — flexible and incremental, i.e. rebuilds only what changed). A **dependency** is an external library your project uses; build tools download the right versions for you. Both read `JAVA_HOME` and support *toolchains* to pin a JDK version. Details: Chapter 18.

### Recommended starting workflow

1. Chapters 1–3: plain `javac`/`java`, `java File.java` and `jshell` — you learn what the tools really do.
2. From Chapter 5 on (multiple classes, packages): an IDE.
3. Anything with dependencies: Maven or Gradle.

### Coding conventions (used throughout this book)

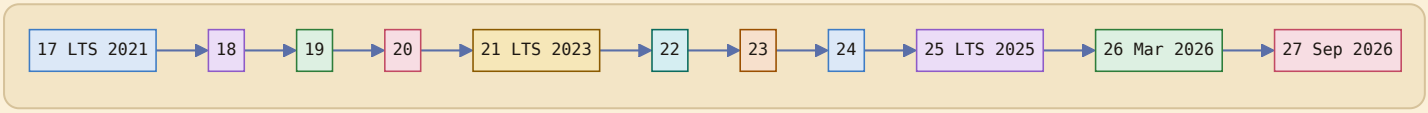
| Kind | Convention | Example |
|---|---|---|
| Class, interface, record, enum | `UpperCamelCase` noun | `OrderService`, `Shape` |
| Method, variable | `lowerCamelCase`; methods are verbs | `computeTotal()`, `itemCount` |
| Constant (`static final`) | `UPPER_SNAKE_CASE` | `MAX_RETRIES` |
| Package | all lowercase, reversed domain | `com.example.billing` |
| Indentation | 4 spaces; opening brace on the same line | `if (x) {` |

---

## 1.12 Java releases, LTS and JEPs

### How the release model works

- A **feature release** ships every six months: **March** and **September** (since Java 10, 2018). Version numbers are just counters: 25, 26, 27…
- Each feature release gets a couple of quarterly **update releases** (e.g. 25.0.1, 25.0.2) from Oracle's OpenJDK builds; vendors then continue patching the **LTS** ones.
- **Long-Term Support (LTS)** releases are the ones vendors support for years: **8, 11, 17, 21, 25** — every two years since 17. The next is expected in September 2027 (Java 29). Oracle offers at least eight years of support for an LTS (five years Premier + three Extended).
- Non-LTS releases (22, 23, 24, 26, 27…) are production-quality but supported only until the next release. Use them to try features early.




### JEPs and feature maturity

New features are proposed in **JEPs** (JDK Enhancement Proposals, openjdk.org/jeps) — numbered documents such as *JEP 512: Compact Source Files and Instance Main Methods*. A feature often goes through stages:

| Stage | Meaning | How to use it |
|---|---|---|
| **Incubator** | An experimental API in a `jdk.incubator.*` module (e.g. the Vector API, still incubating in 27) | `--add-modules jdk.incubator.vector` |
| **Preview** | A complete language/VM/API feature that may still change | `javac --release 25 --enable-preview …` and `java --enable-preview …` |
| **Final / permanent** | Standard, stable part of Java | Just use it |
| **Withdrawn** | Removed before becoming final (e.g. *string templates*, previewed in 21–22, dropped in 23) | — |

This book marks every modern feature with the version that **finalised** it (e.g. *records — Java 16*, *pattern matching for* `switch` *— Java 21*). A full tour is in Chapter 17.

### Which version should I use?

| Situation | Pick |
|---|---|
| New project, or learning | **Java 25 LTS** |
| Existing Java 17/21 service | Stay on it while supported; plan an upgrade to 25 (mostly smooth — run tests with the new JDK first) |
| Legacy Java 8 | Upgrade path 8 → 11 → 17 → 21/25 needs attention to removed Java EE modules and strong encapsulation of JDK internals (your code may no longer reach the JDK's private implementation classes) |
| Library you publish | Compile with `--release` of the *oldest* Java you support |

#### In the interview

**Q: What is an LTS release and why does it matter?** A feature release that vendors support with updates for many years (8, 11, 17, 21, 25); companies standardise on them so they are not forced to upgrade every six months.

- *How often does Java release?* Every six months (March/September); LTS every two years.
- *What's a preview feature?* A finished-but-not-final feature you must opt into with `--enable-preview`; it may change or be withdrawn.
- *Name three Java 21 features.* Virtual threads, pattern matching for `switch`, record patterns, sequenced collections.

#### Self-check

1. Is Java 26 an LTS? → No. 25 is the current LTS; 29 is the next expected.
2. What do you need to compile a preview feature on JDK 25? → `--release 25 --enable-preview` on both `javac` and `java`.
3. Where are new Java features proposed? → In JEPs on openjdk.org.

---

## Chapter TL;DR

- **Java** = statically typed, object-oriented language + portable runtime. Current LTS: **Java 25** (Sept 2025); releases every March/September, LTS every two years.
- **JDK ⊃ runtime (JRE) ⊃ JVM.** Install a JDK; ship a `jlink` runtime or container image.
- `javac` compiles `.java` → `.class` bytecode (major version 69 = Java 25); `java` runs it on the JVM. `--release N` targets an older Java safely.

- The JVM **loads lazily, verifies, interprets, then JIT-compiles** hot code (C1 → C2) using profiles; AOT caches (JDK 24–25) and native images cut warm-up.
- **Stack** holds frames and locals; **heap** holds objects; the **GC** frees unreachable objects. Default G1; ZGC/Shenandoah for low latency.
- **jshell** for experiments; `java File.java` to run sources directly (multi-file since 22); **compact source files +** `void main()` **+** `IO` are final in **Java 25**.
- Exam-trap depth on this chapter's basics: OCP Chapter 1. Next: Chapter 2 — Basics: types, variables and operators.